Relative effects of seagrass canopy, wave coupling and bed mobility on Lagrangian transport in a shallow lagoon

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Abstract

Coastal lagoon models are routinely validated against water level, although habitat connectivity and dispersal depend on transport. This study asks which modelled processes control transport in a shallow vegetated lagoon and whether water level can identify them. A three-dimensional Delft3D Flexible Mesh model of the Stagnone di Marsala, a micro-tidal Sicilian lagoon of sub-metre mean depth colonised by *Posidonia oceanica*, was coupled to SWAN over nine days of July 2025. Eight configurations form a complete factorial in wave coupling, bed mobility and seagrass canopy, with the canopy parameterised from published measurements and distributed by satellite classification, and each was scored against three tide gauges and 35 GPS surface drifters withheld from every model decision. On the fixed bed, Liu–Weisberg skill ranges from 0.597 to 0.739 and divides on the canopy, which raises it by 0.120 and 0.114 with both bootstrap intervals clear of zero, against -0.022 and -0.028 for wave coupling. Water level separates the canopy treatments only at the gauge inside the vegetated basin, by 26 and 27 mm in anomaly RMSE against at most 4 mm near the inlets. With waves, the uncalibrated mobile bed brings a configuration without canopy to the skill of the vegetated ones through a bathymetry reshaped by implausible erosion, a compensation that the interior gauge exposes and the drifters do not. The canopy is therefore the first-order control on transport here, and leaving it out lets errors elsewhere pass as process effects. These results rest on one nine-day window and a stratified initial state. Transport prediction in vegetated lagoons requires canopy drag resolved rather than absorbed into bed roughness, and published measurements with satellite mapping suffice to specify it without tuning.

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1. Introduction

Coastal lagoons are among the most productive coastal environments on Earth. Their shallow, semi-enclosed geometry concentrates fisheries and biodiversity along roughly 13% of the global shoreline (Carrasco et al., 2016; Kjerfve, 1994), while making these systems acutely sensitive to the atmospheric and tidal forcing that drives water exchange and renewal. In the Mediterranean basin, where tidal amplitudes rarely exceed 0.3 m, the balance among tidal oscillations, wind stress, and seiche phenomena varies considerably among systems, yet wind consistently emerges as a significant driver of horizontal exchange and a key modulator of the residence times and salinity gradients that condition benthic habitat (Niedda and Greppi, 2007; Umgiesser et al., 2014). The presence of dense submerged vegetation, principally *Posidonia oceanica* (L.) Delile and *Cymodocea nodosa* (Ucria) Ascherson along Mediterranean shores, introduces a further layer of complexity. Seagrass canopies attenuate incident wave energy (Infantes et al., 2012) and extract momentum from the mean flow through canopy drag, modifying near-bed turbulence and the vertical velocity structure (Nepf, 2012). Global seagrass extent has fallen in all seven bioregions over the long record, by 19.1% of the surveyed meadow area since 1880 (Waycott et al., 2009; Dunic et al., 2021), and the steepest recent trajectories track coastal development and water quality rather than any single driver (Turschwell et al., 2021). How circulation controls residence times, dispersal, and habitat connectivity in vegetated coastal systems is therefore a question for management as much as for science. Addressing it requires circulation models that accurately reproduce transport pathways rather than water levels alone, and this distinction makes Lagrangian validation essential. By tracking the cumulative separation between simulated and observed drifter trajectories over time, the Lagrangian skill score provides a direct and discriminating measure of model accuracy that conventional scalar metrics cannot replicate (Liu and Weisberg, 2011; Révelard et al., 2021).

The *Stagnone di Marsala*, on the north-western coast of Sicily, offers a natural laboratory for these questions. Its mean water depth of approximately 0.95 m places it firmly in the class

of very shallow coastal systems in which the depth-averaging assumption is, a priori, difficult to justify, and its floor carries meadows of *P. oceanica* and *C. nodosa* dense enough to be a candidate first-order control on the flow (De Marchis et al., 2012; Ciraolo et al., 2009a). Satellite monitoring over 2003–2024 has documented a 75% reduction in the extent of the *P. oceanica* atoll and cordon formations (Maltese et al., 2025), so the vegetated state a model of this system must represent is itself changing. The physical setting is described in Section 2.

The absence of riverine input leaves circulation driven almost exclusively by tidal oscillation and wind stress. Field experiments and 3D numerical modelling conducted by De Marchis et al. (2012) during a July 2006 campaign showed that tidal action dominates north–south exchange, while wind is the controlling agent of east–west currents and generates pronounced vertical recirculation structures entirely absent in tide-only simulations. This established that vertical velocity shear between wind-driven surface drift and near-bed return flow is quantitatively significant even at sub-metre depths, and that the tendency towards hypersaline conditions in summer (La Loggia et al., 2004) adds a baroclinic component that reinforces stratification in the vertical.

Prior numerical investigations of the Stagnone have employed both depth-averaged and three-dimensional approaches, each appropriate to the research objective pursued. La Loggia et al. (2004) and Di Marca et al. (2009) applied 2D depth-averaged models to characterise tidal flushing, residence times, and turbidity dynamics, yielding useful insight into the basin-scale water budget and the role of biological productivity in light attenuation. Ciraolo et al. (2009a) demonstrated through particle tracking the sensitivity of simulated trajectories to the assumed velocity field, highlighting the importance of current accuracy for transport prediction. De Marchis et al. (2012) showed that resolving the vertical dimension is necessary to reproduce wind-driven recirculation. Their 3D non-hydrostatic finite-volume model, validated against simultaneous velocity and water-level measurements at multiple stations, achieved substantially better agreement with observed east–west currents than any depth-averaged simulation. More recently, Ingrassia et al. (2024) returned to a 2D unstructured-mesh framework (MIKE 21) and conducted systematic sensitivity tests of bed friction, varying the Gauckler–Strickler coefficient uniformly across the lagoon for different vegetation scenarios. The value $K_s = 20$ m$^{1/3}$ s^{-1} was identified as most representative for the vegetated basin, yielding Nash–Sutcliffe efficiencies

up to 0.92 for water level. The authors noted that the 2D approximation remains adequate for large-scale water-level budgeting while acknowledging reduced predictive skill for horizontal velocity, particularly in the northern, shallower sub-basin.

Building on these foundations, the present study addresses two aspects that have not yet been incorporated jointly into a Stagnone circulation model. The first is wave–current coupling. Spectral ocean waves generated in the western Sicilian channel can penetrate both inlets and interact with the mean flow, altering near-bed turbulence and the effective drag experienced by the seagrass canopy. Coupled wave and flow modelling in comparably shallow Mediterranean basins has shown this to be non-negligible. In the Venice Lagoon Carniello et al. (2011) found the wave-induced contribution to bed shear stress to interact non-linearly with the current-induced contribution, so that neither can be obtained from the other. In the Mar Menor, a restricted micro-tidal lagoon of comparable depth, Morote-Sánchez et al. (2025) found locally generated waves to produce near-bed orbital velocities sufficient to mobilise bottom sediment over most of the basin. Neither process has been represented in previous Stagnone simulations. The second is the combined use of a spatially distributed canopy drag, its parameters taken from published measurement and its distribution from satellite classification rather than assumed uniform, and of Lagrangian drifter observations as a validation target. Together, these extensions allow the relative contributions of three-dimensional dynamics, wave forcing, and seagrass heterogeneity to transport skill to be quantified within a controlled process-attribution framework.

The study applies the three-dimensional unstructured-mesh hydrodynamic solver Delft3D Flexible Mesh (D-Flow FM; Kernkamp et al., 2011) coupled online with the spectral wave model SWAN (Booij et al., 1999), driven by ERA5 reanalysis meteorological forcing and CMEMS Mediterranean physical reanalysis boundary conditions, over a nine-day period in July 2025. The seagrass canopy is represented as a three-dimensional drag field resolved per sigma layer, with its parameters taken from published flume and large-scale measurements of the same species and its spatial distribution from a supervised random-forest classification of PlanetScope multispectral imagery acquired in August 2023, following the mapping methodology of Maltese et al. (2025). Model performance is assessed against in-situ water level time series at three lagoon stations and an offshore tide gauge, and against GPS surface

drifter trajectories from a field campaign concurrent with the simulation period, using the Liu–Weisberg Lagrangian skill score and endpoint proximity as metrics. A complete $2 \times 2 \times 2$ factorial in wave coupling, seagrass canopy and morphodynamic bed update attributes differences in Lagrangian skill to individual physical mechanisms and to their interactions, with the attribution made on its fixed-bed half and the mobile-bed half kept as a sensitivity test (Section 3.6).

The study pursues three objectives. The first is to rank the three processes by their effect on Lagrangian transport skill. The second is to establish whether water level, the quantity against which shallow lagoon models are conventionally validated, can identify that ranking. The third is to measure how far the apparent effect of each process depends on how the other two are represented. The second and third are not specific to this basin, concerning what a class of validation data can certify and how a factorial behaves when a dominant factor is left out of it.

2. Study Area

The *Stagnone di Marsala* is located on the north-western coast of Sicily, sheltered from the western Sicilian channel by Isola Grande, an elongated barrier island running parallel to the mainland coast (Figure 1). The enclosed basin extends approximately 11 km in the north–south direction and 2.5 km east–west, enclosing a total surface area of roughly 2200 ha (De Marchis et al., 2012). Two geomorphologically distinct sub-basins are separated by a *Posidonia oceanica* barrier reef (La Loggia et al., 2004): the northern sub-basin, with a mean depth of approximately 1 m and extensive intertidal flats, and the deeper southern sub-basin, with a mean depth of approximately 2 m and a maximum of about 3 m near the main inlet. Three islands punctuate the interior, the Phoenician archaeological site of Mothia in the south-central basin and the smaller islands of Santa Maria and Scuola in the central reaches (Ciraolo et al., 2009b). Salt pans (*saline*) occupy both margins of the lagoon, with extensive historic saltworks on the western barrier island (Isola Grande) and additional pans along the eastern mainland shore in the area facing Mothia (Basso et al., 2008). They constitute a distinctive shallow intertidal habitat within the lagoon system.

Hydrodynamic exchange with the open sea occurs through two morphologically distinct

inlets. The northern inlet (approximately 400 m wide, 0.30–0.40 m deep) is a highly restricted passage whose tidal prism is severely limited by its shallow sill. A historically dredged channel (20 m wide, 1 m deep) partially compensates this restriction (Ciraolo et al., 2009b). The wide southern inlet (approximately 2900 m wide, 1.0–1.5 m deep) provides the primary pathway for mass transport and water mixing between the lagoon and the open sea (Ciraolo et al., 2009b). The tidal regime is microtidal, with free-surface amplitudes of approximately 0.30 m dominated by semi-diurnal harmonics (La Loggia et al., 2004). In the absence of riverine input, weak tidal exchange combined with high summer evaporation drives a pronounced salinity gradient. Annual lagoon-wide salinity spans 33–46 psu (Basso et al., 2008), with peak summer values approaching 48 psu in the more confined northern sub-basin, contrasting with near-marine conditions near the southern inlet (Mancuso et al., 2023). Wind forcing dominates east–west circulation within the basin, while tidal exchange controls the north–south component (Ciraolo et al., 2009b; De Marchis et al., 2012).

The seagrass mosaic covers the majority of the lagoon floor. *Posidonia oceanica* occupies the central and southern portions of the basin, forming both continuous meadows and characteristic atoll patterns (10–20 m in diameter) that grade into reef formations oriented along the north–south axis. *Cymodocea nodosa* prevails in the shallower northern reaches, with *Caulerpa prolifera* replacing *P. oceanica* locally where hydrodynamic and sediment conditions deteriorate (La Loggia et al., 2004; Ciraolo et al., 2009b). In Mediterranean meadows of this type, *P. oceanica* areal density reaches 1000–1200 plants m^{-2} (Ciraolo et al., 2009a). The spatial distribution of submerged vegetation was first mapped from airborne hyperspectral imagery by Ciraolo et al. (2006a). The roughness parametrisation in this study rests on an independent classification of PlanetScope imagery for the 2023 epoch, described in Section 3.4. The lagoon is designated both as a *Riserva Naturale Orientata* and as a Natura 2000 site, reflecting the conservation importance of these meadows. The extent of the atoll and cordon formations nevertheless fell by approximately 75% between 2003 and 2024 (Maltese et al., 2025).

The observational dataset used in this study was collected during July 2025 (Figure 1). Water level was recorded at three stations, BocaNord (BN) at the northern inlet, BocaSud (BS) near the southern inlet, and AltaVilaEst (AE) on the eastern flank of the central basin.

Wind speed and direction and atmospheric pressure were measured at two stations, AE (which also carries a water-level gauge and a water temperature sensor) and Mulino, a dedicated meteorological station located near the salt pans in the central-northern area. Water levels at BN, BS, and AE serve as the primary validation targets for the hydrodynamic model (Section 3.5). Open-sea boundary conditions are prescribed from CMEMS Mediterranean physical reanalysis products (Section 3.3). Offshore model performance is assessed against the tidal gauge on Marettimo island (JRC TAD 658), approximately 35 km to the west, which provides a free-surface reference uninfluenced by lagoonal dynamics. GPS-tracked surface drifters were released in the lagoon interior over a sub-period coinciding with the simulation window, and their trajectories form the basis of the Lagrangian validation (Section 3.5).

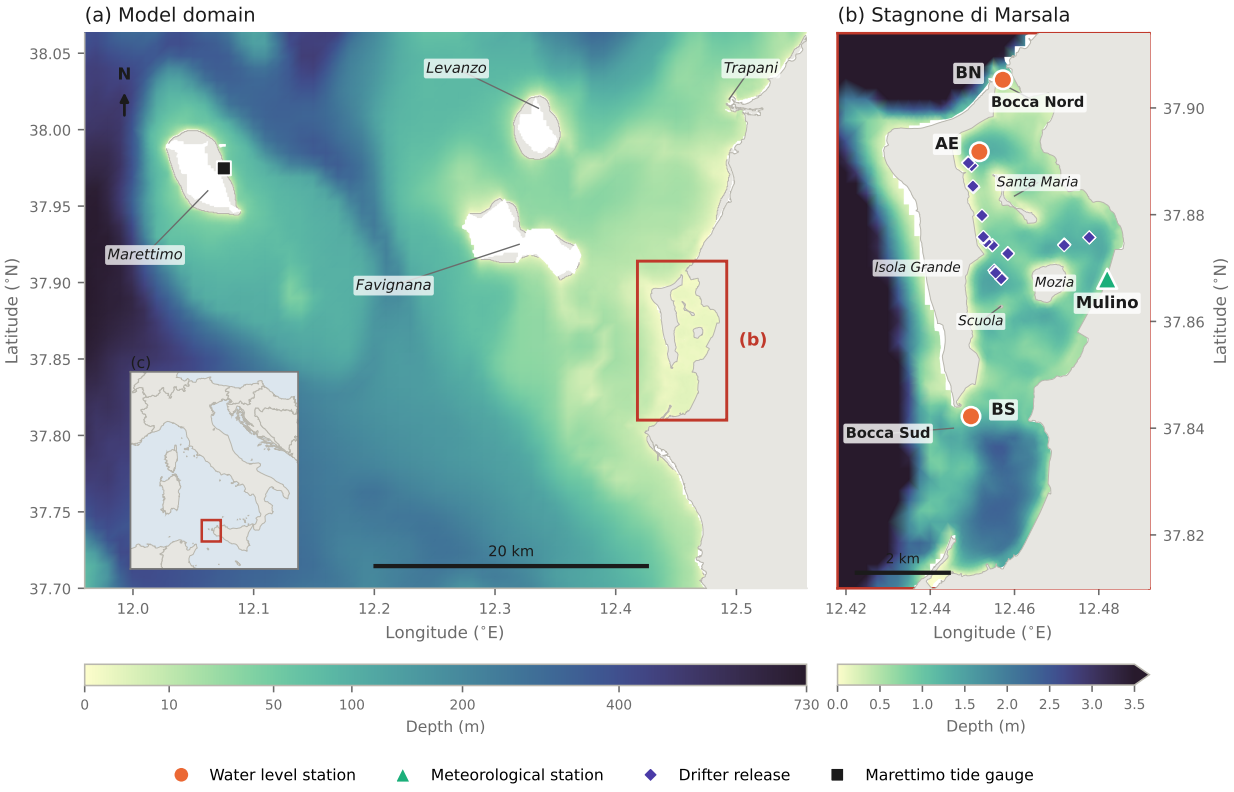


Figure 1: The study area. (a) The modelled domain, covering western Sicily, the Egadi archipelago and the adjacent shelf, with the Marettimo tide gauge used for offshore validation. The red box gives the extent of panel (b). (b) The Stagnone di Marsala, with the water level stations (BocaNord, AltaVilaEst, BocaSud), the Mulino meteorological station, and the 35 GPS drifter release positions of the twelve July 2025 deployments. (c) The domain within Italy. Colour gives depth below the reference level on the model mesh, on separate scales for (a) and (b).

3. Data and Methods

3.1. Hydrodynamic model

Circulation was simulated with D-Flow Flexible Mesh (D-Flow FM, kernel 1.2.184) from the Delft3D FM Suite 2026.01, an unstructured finite-volume solver for the three-dimensional shallow-water equations (Kernkamp et al., 2011). Momentum advection used the Perot scheme with a monotone central limiter on the cell-centre velocity, and the same limiter was applied to the transport of salinity and temperature. The water-level system was solved with a preconditioned conjugate-gradient method. Bed levels were specified at cell centres from the bathymetry file.

The computational mesh comprised 21,188 nodes and 25,212 faces, 14,969 of them quadrilateral and 10,243 triangular, and covered the lagoon together with an offshore extension reaching the western Sicilian channel, spanning 11.96–12.56°E and 37.70–38.06°N in geographic (WGS84) coordinates. Taking the distance between the centres of adjacent wet cells as the measure of resolution, the mesh grades from about 880 m in the open offshore to about 60 m in the lagoon and in the bands that follow the Egadi and mainland shorelines. The constraint the grading answers is the width of the inlets, which carry the exchange that sets the interior response. At the 61 m median spacing around the northern inlet, about six cells span its 400 m width, and about forty span the 2900 m southern inlet, while the 20 m dredged channel through the northern sill falls below the mesh scale and enters only through the cell-centre bathymetry.

The vertical was discretised with 10 sigma layers of uniform thickness, each a tenth of the local water depth, which at the mean depth of 0.95 m gives layers of about 0.095 m. The count answers two features of the problem, since resolving the wind-driven surface drift over a near-bed return flow reported by De Marchis et al. (2012) requires several layers through a sub-metre column, and the 0.40 m canopy of Section 3.4 spans the lowest four of the ten layers and part of the fifth at that depth, so that its drag is distributed over the vertical rather than collapsed into a single near-bed term. The sensitivity of the results to the horizontal and the vertical discretisation was tested separately (Section 3.6).

Turbulence closure used the k–ε model with background vertical eddy viscosity and diffusivity of 10^{-4} $\mathrm{m^2\,s^{-1}}$ and uniform horizontal values of 0.1 $\mathrm{m^2\,s^{-1}}$. Density was computed

from the UNESCO equation of state, with salinity and temperature both transported as active constituents so that the summer hypersaline gradient contributes baroclinically to the momentum balance. Bed roughness was prescribed as a Manning coefficient, uniform at 0.023 s $m^{-1/3}$ in every member, which is the default of the solver. The value stands for the bare substrate alone and was not calibrated, being held uniform by design so that the seagrass canopy, carried separately as a three-dimensional drag field (Section 3.4), is the only spatially variable resistance in the ensemble. Wind drag followed the Charnock formulation with the Charnock coefficient taken from the ERA5 field (Section 3.3) rather than held constant.

The simulation covered nine days, from 1 July 2025 00:00 to 10 July 2025 00:00 UTC. External forcing was updated every 300 s, shorter than the interval of every forcing record used, so the update does not set the time resolution of the forcing. The computational time step was adaptive under a Courant limit of 0.7 and capped at 30 s in the members coupled to the wave model and at 15 s in the uncoupled ones, at least an order of magnitude below the forcing update so that no forcing interval is crossed in a single step, and constrained to grow by no more than a factor 1.1 between successive steps, which keeps the step from recovering abruptly after the wetting and drying events on the intertidal flats where the Courant limit binds. Initial conditions were ramped over the first hour.

The domain was decomposed into eight partitions and run under MPI. Eight partitions leave about 3000 cells each, above the few thousand below which the communication cost of the parallel solver begins to dominate for meshes of this size (Mogé et al., 2019). All members were run on the same 48-core Linux server with the same decomposition, where a coupled nine-day member completed in four and a half to six hours of wall-clock time and an uncoupled member in under three.

3.2. Wave model and coupling

Waves were computed with SWAN (Booij et al., 1999) through the Delft3D-WAVE module, on two nested curvilinear grids. The outer grid, 69×54 points at approximately 800 m, covered the full D-Flow FM domain and took its boundary spectra from the western, southern and northern edges. It supplied boundary conditions to an inner grid of 121×221 points at approximately 90 m east–west and 110 m north–south over the lagoon and its inlets. The outer grid matches the 880 m background of the hydrodynamic mesh offshore. The inner grid

is about half as fine as the 60 m flow mesh inside the lagoon, spanning the 400 m northern inlet with about five cells against the flow mesh's six, so the wave field is the coarser of the two representations of the inlets and is interpolated onto a finer flow mesh.

The spectral space was discretised into 24 directions over the full circle, a resolution of 15°, and 24 frequencies logarithmically distributed between 0.05 and 1.0 Hz. The range spans periods from 1 to 20 s, which brackets both the swell entering through the inlets and the short waves generated within the basin, whose fetch-limited peak period over the 2.5 km cross-basin fetch is of the order of 2 s at the wind speeds observed.

Third-generation source terms were used, with Komen whitecapping, quadruplet wave–wave interactions, refraction, and frequency shifting all active, which is the standard set for a wind-sea dominated application. Triad interactions were left inactive. They redistribute energy within the spectrum in very shallow water, and the quantity exchanged with the flow solver is the depth-integrated dissipation rather than the spectral shape, so their omission bears on the distribution of energy over frequency more than on the momentum passed to the flow. Wave setup was also left inactive, since the free surface is computed by the flow solver, which already receives the wave forcing, and activating it in the wave model would apply that contribution twice. Depth-induced breaking followed the formulation of Battjes and Janssen (1978) with $\alpha = 1.0$ and $\gamma = 0.73$, the mean value Battjes and Stive (1985) obtained over laboratory and field data, and bottom friction used the JONSWAP formulation with a coefficient of 0.067 $\mathrm{m^2\,s^{-3}}$, the value for wind sea rather than the lower value appropriate to swell (Bouws and Komen, 1983). A minimum depth of 0.05 m excluded the thin films on the intertidal flats, where the spectral formulation carries no meaning, from the wave computation. Wave forces were passed to the flow solver as three-dimensional dissipation.

Flow and waves were coupled online through the DIMR framework, exchanging fields every 3600 s. The interval was set by the time scales the coupling has to resolve, the 12.4 h tide and the diurnal sea breeze, and by cost, the wave model accounting for about four fifths of the wall time of a coupled run. It bounds the time resolution of the wave field, and since the drifter deployments of Section 3.5 last between 0.4 and 7.2 h, the shortest of them are advected through a single wave field.

Within each exchange the wave model ran in stationary mode, so the time variation of the

wave field followed from the re-evaluated water levels, currents, wind, and boundary spectra rather than from spectral propagation in time. The assumption this rests on is that the wave field equilibrates with its forcing faster than the forcing changes. Inside the lagoon, where the fetch is at most 2.5 km across the basin and the depth is of order 1 m, wave energy crosses the generation area in about fifteen minutes at the shallow-water group velocity, well within the hourly exchange. Offshore the crossing time is longer, but the sea state there is prescribed from the boundary spectra rather than generated inside the domain. Each stationary computation was iterated until the significant wave height and mean period changed by less than 2% at 98% of the wet points, to a maximum of fifteen iterations. Communication files were written in classic NetCDF (`ncFormat = 3`), which was required for the wave module to re-open the communication file reliably across the coupled run.

3.3. Boundary and atmospheric forcing

Open-sea boundaries were forced with the CMEMS Mediterranean Sea physical reanalysis. Water level, salinity, and temperature were prescribed along the offshore boundary, together with a depth-resolved advection velocity (`uxuyadvectionvelocitybnd`). Prescribing the advection velocity was necessary for stability in the deep offshore cells, where a water-level-only boundary admitted spurious accelerations.

The CMEMS sea surface height field is an anomaly about the model rest state rather than a geodetic level, so it cannot be imposed directly as a free surface without a reference correction. A per-node offset was therefore added to the water-level boundary, anchored on the tide gauge at Marettimo island (JRC TAD 658) and distributed along the boundary using the CMEMS mean dynamic topography. The anchor is the mean difference between the gauge record and the CMEMS field interpolated to the gauge, both reduced to daily means, over 384 aligned days spanning January 2025 to January 2026. It amounts to +0.449 m, of which the gauge contributes +0.091 m and the CMEMS rest-state reference the remaining −0.358 m. The monthly means of that difference range from +0.361 to +0.505 m, so a seasonal residual of about 0.07 m either side of the annual value is not removed by a constant anchor, which bounds the accuracy of the reference level the model can be held to.

Atmospheric forcing combined ERA5 reanalysis with in-situ observation. Wind components, mean sea level pressure, and the Charnock parameter were taken from ERA5, and the

wind field was blended with the record from the AltaVilaEst station. Earlier configurations that additionally blended a second in-situ station produced a systematic rotation of the wind vector along the eastern lagoon margin, so the AltaVilaEst record alone was retained. Evaporation was imposed through the rainfall quantity with a negative sign, which is the convention D-Flow FM expects, with both the rainfall and evaporation switches enabled in the model definition. Interior salinity and temperature were nudged towards the CMEMS fields with a relaxation time of 8.64×10^5 s, long enough to leave the lagoon interior essentially free-running while constraining the offshore extension.

Temperature and salinity outside the lagoon were initialised with the depth profiles of the CMEMS nudging field at the start time, applied over the domain, from 25.8 °C and 37.5 at the surface to 16.2 °C and 38.3 below 260 m. The hypersaline value of 42 was prescribed over the lagoon, whose temperature takes the surface value of the profile.

3.4. Seagrass canopy parametrisation

The canopy was represented as a three-dimensional drag field, resolved per sigma layer over the vertical extent the plants occupy. A cell whose canopy fills two fifths of the water column carries drag in the layers within that fraction and none above it. Drag per unit volume follows the standard quadratic form (Nepf, 2012), with a drag coefficient C_D, a stem diameter D, an areal stem density m and a canopy height h_v.

The alternative available in this solver is the trachytope framework, in which vegetation enters as a composite bed roughness following Baptist et al. (2007), and which the earlier depth-averaged work on this lagoon uses. It was not adopted here. A roughness term acts at the bed, whereas the quantity the factorial has to separate is a momentum sink distributed through the part of the column the plants occupy, which is the distinction between a canopy that shelters the bed and one that resists the flow over four tenths of the depth.

Table 1 gives the values and their provenance. None was fitted to any observation used to evaluate the model. Densities are expressed per leaf rather than per shoot, since it is the leaf that presents the frontal area and *P. oceanica* carries about six leaves per shoot (Ciraolo et al., 2006b). The canopy height is the vertical extent the drag occupies, which for a plant whose leaves lie over is shorter than the leaf. The flume measurements of Ciraolo et al. (2006b) give deflected canopy thickness between 0.2 and 0.6 of the water depth, and the large-scale

experiments of Prinos et al. (2010) adopt submergence ratios of 0.323, 0.423 and 0.500, so 0.40 m was taken for a basin of mean depth 0.95 m. The drag coefficient is the rigid-cylinder value.

Fractional cover from the classification multiplies the areal density and nothing else, height and diameter being properties of the plant rather than of how much bed it covers. Median cover is 0.685 for *P. oceanica* and 0.181 for *C. nodosa*.

Bed friction is a uniform Manning coefficient of 0.023 s $m^{-1/3}$ in every member of the ensemble, vegetated or not, so the canopy is the only spatially variable resistance in the design and the contrast between the two arms is not confounded by a difference in bare-substrate roughness.

Table 1: Canopy parameters. *P. oceanica* values are from Ciraolo et al. (2006b), except the canopy height, which is from Prinos et al. (2010). *C. nodosa* values are from Morris et al. (2008), with the shoot density also bracketed by Cancemi et al. (2002).

Parameter	*P. oceanica*	*C. nodosa*
Shoot density (m^{-2})	500	800
Leaves per shoot	6	3
Leaf width D (m)	0.010	0.004
Canopy height h_v (m)	0.40	0.15
Leaf density (kg m^{-3})	900	900
Drag coefficient C_D	0.8	0.8

The *C. nodosa* values come from a flume study of that species collected in a sheltered tidal lagoon (Morris et al., 2008), which measured leaf width of 3.8 ± 0.4 mm, 3.5 ± 1.1 leaves per shoot and an undeflected canopy height of 0.17 ± 0.02 m. Its shoot density of 520 ± 69 m^{-2} and the Mediterranean field range of 925 to 1925 m^{-2} (Cancemi et al., 2002) bracket the value used. Leaf density was carried across from *P. oceanica*, no measurement of the second species being available, and it governs only the bending response. The class occupies 22.5% of the classified links against 48.6% for *P. oceanica*, so at the cover fractions above it carries about 1.5% of the total canopy frontal area.

The spatial assignment of classes came from a supervised random-forest classification of PlanetScope SuperDove imagery acquired in August 2023 at 3 m ground sampling distance. The classification follows the approach of Maltese et al. (2025) in applying depth-invariant indices to remove the water column signal before supervised classification, but it is an

independent product, a random forest on 3 m PlanetScope scenes in place of their gradient-boosted classifier on 2 m WorldView, QuickBird and Pleiades imagery. Classified pixels were transferred to the mesh with a 20 m search radius, and a flow link straddling more than one class carries each class weighted by the area it occupies (Figure 2). Of the 7345 links inside the lagoon, 48.6% are dominated by *P. oceanica*, 22.5% by *C. nodosa*, 25.6% by bare sand and 3.3% by rock. The resulting roughness link file must be written in the same geographic coordinate system as the mesh, because a projected file is accepted without error while assigning no links at all.

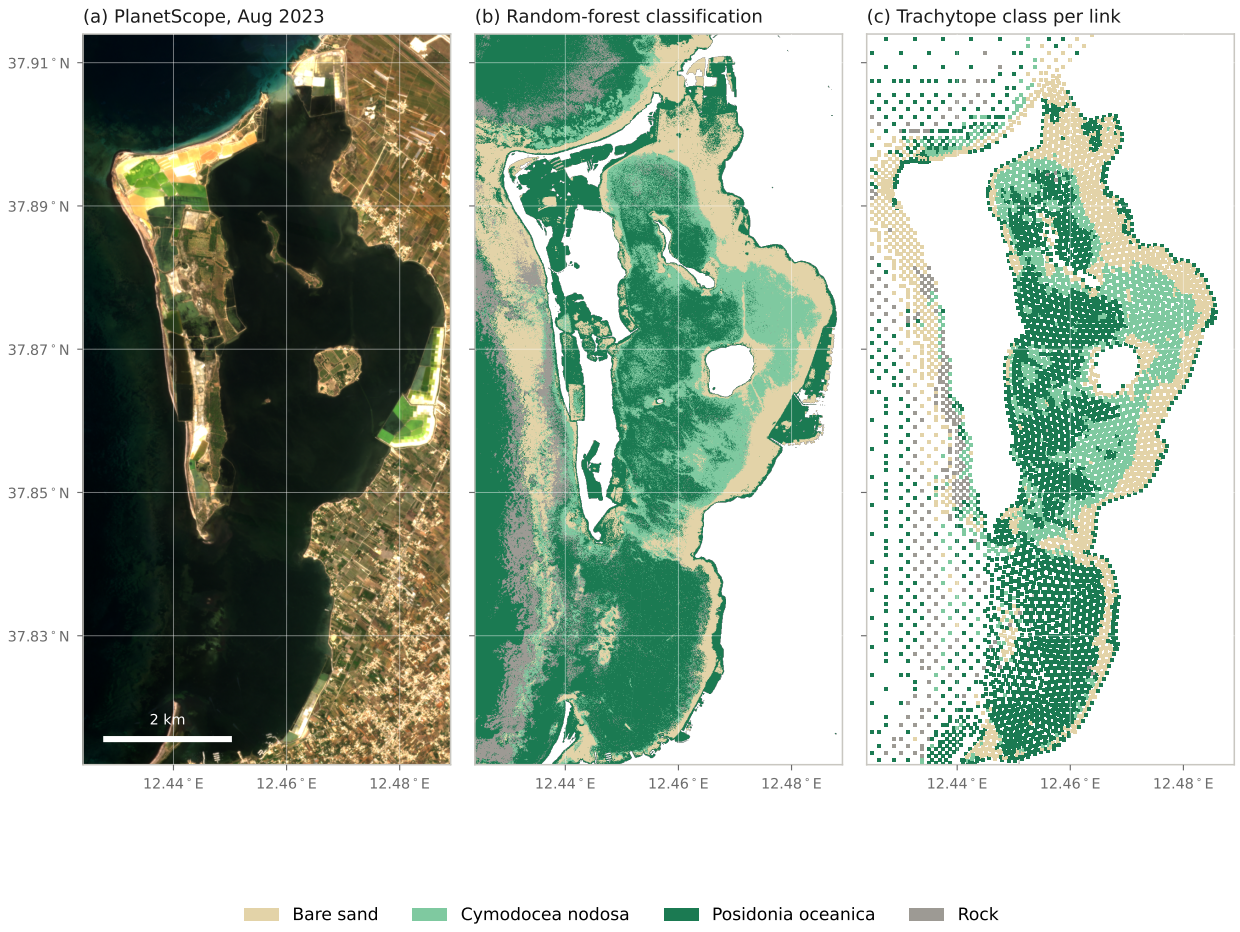


Figure 2: The seagrass map from imagery to model input. (a) PlanetScope SuperDove August 2023 median composite. (b) The random-forest classification, with the six output classes collapsed to the four seagrass classes. (c) The dominant class at each flow link, from the classification file the model reads, showing the mesh resolution the classification is sampled onto. The classification is computed over the whole scene, but only links inside the lagoon carry a canopy class.

For the vegetated Stagnone in a depth-averaged model, Ingrassia et al. (2024) obtained

a best-fit uniform Gauckler–Strickler coefficient of 20 $\mathrm{m}^{1/3}\ \mathrm{s}^{-1}$, equivalent to a Manning coefficient of 0.050 $\mathrm{s}\ \mathrm{m}^{-1/3}$. That value is a bulk roughness of the vegetated lagoon, and used as the bare-substrate roughness of a model that resolves the canopy separately it would count the canopy twice.

3.5. Observational data and validation metrics

Water level was recorded at ten-minute resolution at three stations, BocaNord at the northern inlet, BocaSud near the southern inlet, and AltaVilaEst on the eastern flank of the central basin. Offshore performance was assessed at the Marettimo tide gauge.

Water-level skill was quantified with the raw root-mean-square error, the bias, and the anomaly root-mean-square error computed after removing the respective means, together with the linear correlation and the ratio of modelled to observed standard deviation. The three error measures are related by $\mathrm{RMSE}^2 = \mathrm{bias}^2 + \mathrm{RMSE}_{\mathrm{anom}}^2$, so reporting them jointly separates a constant reference-level offset from genuine dynamic error in phase and amplitude. The first simulated day was discarded as spin-up, over which the initial condition relaxes and the transients it generates leave the basin.

Lagrangian skill was assessed against GPS-tracked surface drifters released in the lagoon interior during the simulation window. Simulated trajectories were obtained by advecting particles with OpenDrift (Dagestad et al., 2018) through the modelled surface current field, at an advection step of 300 s and with positions written every 600 s. Scoring was restricted, for each drifter, to the interval between its first and last observation, since the advection continues to the end of the forcing while the deployments last between 0.4 and 7.2 h. A wind drift factor of 0.02 was applied to every member. A sweep over this parameter found no single value optimal across deployments, the optimum instead falling as the mean wind speed rose, so a per-member or per-deployment value would have confounded the process attribution with a tuning difference and the one value was held fixed throughout.

Agreement was quantified with the normalised cumulative Lagrangian separation skill score of Liu and Weisberg (2011), which accumulates the separation between modelled and observed positions and normalises it by the distance travelled by the observed drifter, and with the endpoint proximity, the separation between modelled and observed positions at the

end of each deployment. Révelard et al. (2021) discuss the sensitivity of this class of metric to the normalisation choice in coastal applications.

3.6. Numerical experiments

Eight simulations were run over the same nine-day window, from the same initial condition and with identical boundary and atmospheric forcing, so that the differences between them isolate individual model components. All eight share the mesh, and all wrote station output every 600 s and gridded output every 1800 s, so the comparison is not affected by differences in output sampling.

Members are identified by a three-slot code whose slots mark wave coupling, seagrass canopy and mobile bed in that order, carrying `W`, `V` or `M` where the factor is active and a dash where it is not. The eight are `---`, `-V-`, `W--`, `WV-`, `--M`, `-VM`, `W-M` and `WVM`, and the code is used throughout the results.

The design is a complete factorial in three factors, wave coupling, seagrass canopy and bed mobility, and all eight cells were obtained. The process attribution is made on the four fixed-bed members, a two-factor design in wave coupling and canopy, and the four mobile-bed members form a sensitivity arm, for the reason given at the end of this section. Each vegetated member is a three-day segment restarted on 7 July from the state of the bare member of the same cell, so the two members of a pair share the first six days and differ by the canopy over the window in which they are scored. Suspended sediment does not contribute to the density field in the four mobile-bed members, which therefore share that setting, and the fixed-bed members carry no sediment module.

The setting is not a free parameter, since with suspended sediment feeding the density field the two mobile-bed members without waves terminated on a velocity overflow in a deep offshore cell, at the same cell and the same simulated instant to six decimal places in the velocity. In a cell that deep and carrying no erodible thickness, the baroclinic pressure gradient is the only route by which advected sediment can move momentum, and restricting the erodible layer by depth therefore leaves it untouched. Removing that contribution completed both runs and also removed a spurious 2.4 m deposition at 500 m depth, bringing the largest modelled bed-level change in the domain inside the lagoon.

Erodible sediment is confined to depths at which waves can plausibly move it. Available thickness is held at 0.5 m per fraction above 15 m depth and tapered linearly to zero at 20 m, below which mean orbital velocity at the bed is 0.016 $\mathrm{m\,s^{-1}}$, an order of magnitude below the range in which the sand fraction mobilises. Without this restriction the bed erodes over the whole domain, including water hundreds of metres deep where the process has no physical meaning. The restriction bounds erosion and not deposition, so material mobilised in shallow water still settles offshore.

The morphodynamic factor is a sensitivity test rather than a calibrated prediction, the critical shear stress for erosion not having been constrained against a repeated bathymetric survey. Its effect on transport was traced with two segments of the wave-coupled mobile-bed member without canopy, restarted on 7 July from its state on that date, one with the bed updated as in the member and one with the bed held at that state.

The sensitivity to the initial condition, to the numerical discretisation and to the host was tested outside the factorial. The whole ensemble was repeated from a vertically uniform column, at 24 °C and 37.5 outside the lagoon and with the same hypersaline value inside it. Horizontal resolution was varied on the uncoupled fixed-bed configuration without canopy from the uniform start, under a Courant limit of 0.5, with a family of quadtree meshes built in a conformal projection, with the domain, offshore grading and open boundary of the present mesh, one bed surface common to the family and to a run of the present mesh, and a cell edge of 60, 30 and 15 m inside the lagoon and within 500 m of it. Vertical resolution was varied on the same configuration with 10 and 20 uniform layers from the stratified start, with the time step limited also by the vertical fluxes (Deltares, 2025), which the start from a uniform column requires above 10 layers next to the open boundary. The effect of that time-step limit was measured separately at 10 layers. The uncoupled member started from the uniform column was also repeated on four partitions on a second computer, to test the dependence on the host and the decomposition, under the Courant limit of 0.7 used by the ensemble and under 0.5.

4. Results

All eight members are scored on a common interval, 8 to 10 July 2025, which is the intersection of the members' post-spinup spans and which contains the whole drifter record. Members are coded by wave coupling (W), seagrass canopy (V) and mobile bed (M). The attribution rests on the four fixed-bed members, and the mobile-bed arm is reported beside them in the tables and examined in Section 4.6.

4.1. Water level

Table 2 reports water level skill at the three interior stations. Raw error is dominated by a positive bias at BocaNord and AltaVilaEst, between 0.03 and 0.09 m, and by a negative one at BocaSud, between −0.051 and −0.015 m. The bias is a datum and boundary-offset property that all members share and it is not what distinguishes them.

The anomaly RMSE responds to different factors at the three stations (Figure 3b). At BocaSud it varies by 1.6 mm across the four fixed-bed members. At BocaNord it varies by 12 mm and follows wave coupling, falling from 0.039 to 0.027 m without a canopy and from 0.038 to 0.031 m with one. At AltaVilaEst it varies by 32 mm and follows the canopy, the two vegetated members giving 0.021 and 0.028 m with correlations of 0.94 and 0.95 and the two bare members 0.048 and 0.054 m with correlations of 0.79 and 0.81. Modelled-to-observed standard deviation ratios move the same way, from 1.18 and 1.37 without a canopy to 1.01 and 1.18 with one. The mobile-bed members reproduce the values of their fixed-bed counterparts to within 3 mm, with one exception, the bare wave-coupled member at AltaVilaEst, whose anomaly RMSE is 8 mm lower than on the fixed bed and remains 16 mm above that of its vegetated counterpart.

The record itself shows where that error sits (Figure 3a), with the bare configurations overshooting the range of the observed anomaly and running ahead of it, most visibly on the falling limb of 9 July, while the vegetated ones track both the amplitude and the phase. The two arms are separated by more than the spread within either of them over most of the window.

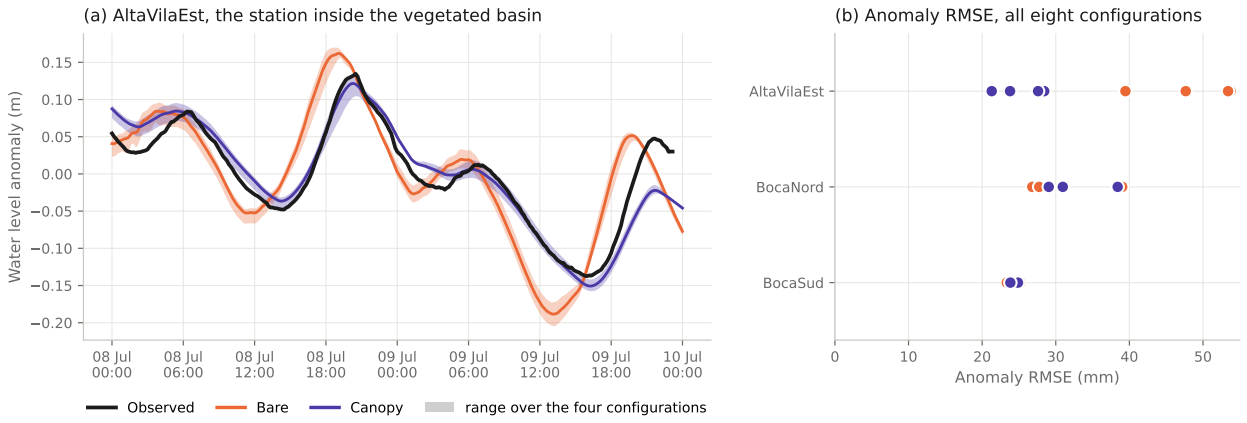


Figure 3: Water level over the scored window. (a) Anomaly at AltaVilaEst, observed against the two arms, each drawn as its mean and the envelope of its four configurations, the mobile-bed ones included. Anomalies rather than raw levels, because the bias is common to all eight members. (b) Anomaly RMSE at the three stations for every configuration, coloured by canopy treatment.

AltaVilaEst is the innermost station, in the shallow northern basin where the meadow is densest and the mean depth is smallest, so it is where a canopy would be expected to matter most, while the two outer stations sit near the inlets where the exchange is set by the boundary.

The vegetated members are three-day segments restarted from the bare member of the same cell, so the separation at AltaVilaEst could in principle reflect the restart rather than the canopy. A segment of the bare wave-coupled mobile-bed member, restarted on 7 July without any other change, reproduces the continuous member to within 1.5 mm in anomaly RMSE at all three stations and to within 0.003 in Liu–Weisberg skill. The restart accounts for none of the difference.

Table 2: Water level skill by member and station, 8–10 July 2025. $\mathrm{RMSE}^2 = \mathrm{bias}^2 + \mathrm{RMSE}^2_{\mathrm{anom}}$. Members are coded by wave coupling (W), seagrass canopy (V) and mobile bed (M). The last row of each station gives the range over the four members of the mobile-bed arm. All values in metres except the dimensionless ratios.

Station	Member	RMSE	Bias	$\mathrm{RMSE}_{\mathrm{anom}}$	r	σ_m/σ_o
BocaNord	`---`	0.052	+0.035	0.039	0.89	1.20
	`-V-`	0.050	+0.033	0.038	0.90	1.22
	`W--`	0.065	+0.059	0.027	0.94	1.08
	`WV-`	0.059	+0.051	0.031	0.92	1.09
	M arm	0.050–0.066	+0.032 to +0.060	0.028–0.039	0.89–0.94	1.12–1.22
BocaSud	`---`	0.050	−0.043	0.025	0.95	1.20
	`-V-`	0.057	−0.051	0.025	0.95	1.16
	`W--`	0.028	−0.015	0.024	0.94	1.06
	`WV-`	0.035	−0.025	0.025	0.93	1.02
	M arm	0.033–0.057	−0.051 to −0.023	0.023–0.025	0.93–0.95	1.05–1.20
AltaVila	`---`	0.076	+0.053	0.054	0.81	1.37
Est	`-V-`	0.049	+0.040	0.028	0.94	1.18
	`W--`	0.097	+0.085	0.048	0.79	1.18
	`WV-`	0.070	+0.067	0.021	0.95	1.01
	M arm	0.050–0.077	+0.042 to +0.062	0.024–0.053	0.81–0.94	1.01–1.36

4.2. Lagrangian transport

Thirty-five drifters across twelve deployments were scored against each member over the interval each drifter was in the water, which ranged from 0.4 to 7.2 hours. Table 3 summarises the metrics and Figure 4 shows their distribution.

On the fixed bed, mean Liu–Weisberg skill spans 0.597 to 0.739 and endpoint separation 327 to 481 m, and the four members divide cleanly on the canopy. The two vegetated members score 0.711 and 0.739 and the two bare ones 0.597 and 0.619, and the two pairs do not overlap. The two components of the trajectory error move together and both improve with the canopy, path length ratio rising from 0.80 and 0.83 without a canopy to 0.88 and 0.94 with one, and mean heading error falling from 11.2 and 10.4° to 8.5 and 8.4°.

Table 3: Lagrangian metrics by ensemble member, 35 drifters over twelve deployments, scored on the interval each drifter was in the water. Path ratio is simulated over observed path length. Heading error is the mean angular difference between simulated and observed step directions and is computed on eleven deployments, deployment 4 having too few positions to form the three step bearings it requires. The lower block is the mobile-bed arm (Section 4.6).

Member	Waves	Canopy	LW skill	Endpoint (m)	Path	Heading
`---`	no	no	0.619	478	0.80	11.2°
`-V-`	no	yes	**0.739**	**327**	0.88	8.5°
`W--`	yes	no	0.597	481	0.83	10.4°
`WV-`	yes	yes	0.711	335	**0.94**	**8.4°**
Mobile bed						
`--M`	no	no	0.614	479	0.80	11.4°
`-VM`	no	yes	0.739	330	0.88	8.5°
`W-M`	yes	no	0.708	304	0.89	7.9°
`WVM`	yes	yes	0.718	307	0.88	8.0°

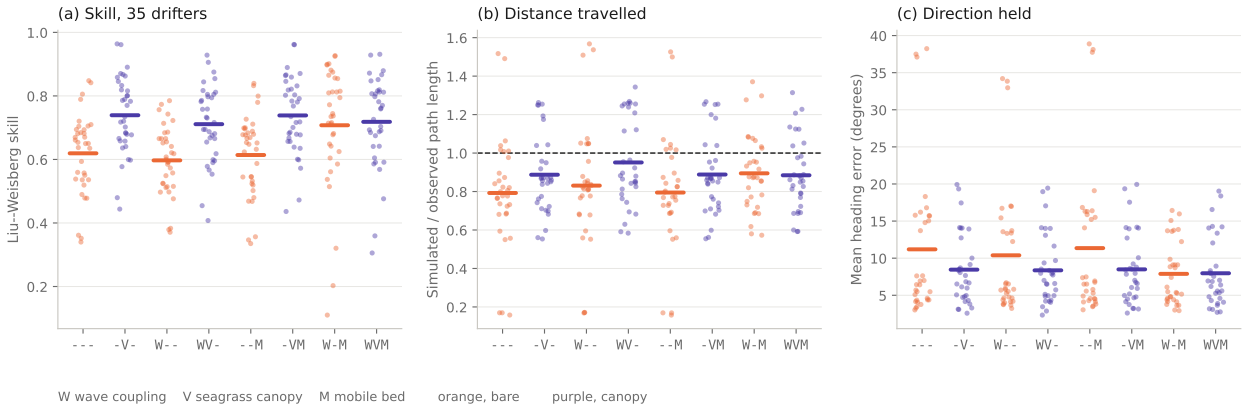


Figure 4: Lagrangian validation, one point per drifter and the bar giving the mean, the four fixed-bed members on the left and the mobile-bed arm on the right. (a) Liu–Weisberg skill. (b) Simulated over observed path length, the dashed line marking equality. (c) Mean angular difference between simulated and observed step directions. Panels (b) and (c) carry 32 drifters rather than 35, deployment 4 having too few positions to form step bearings.

Figure 5 shows the tracks these statistics are computed from, for every deployment and for the wave-coupled fixed-bed pair that differs only in the canopy. In the deployments of the central basin, 1 to 3, the bare particles leave the observed path early and finish short of the observed endpoint while the vegetated ones hold the observed heading. In the southern deployments, 8 to 12, both follow the observed path closely, and the vegetated particles run beyond the observed endpoint in deployments 10 and 11. Deployment 4 is the exception, both configurations leaving to the east and northeast where the observed drifter went east-southeast

over 162 m.

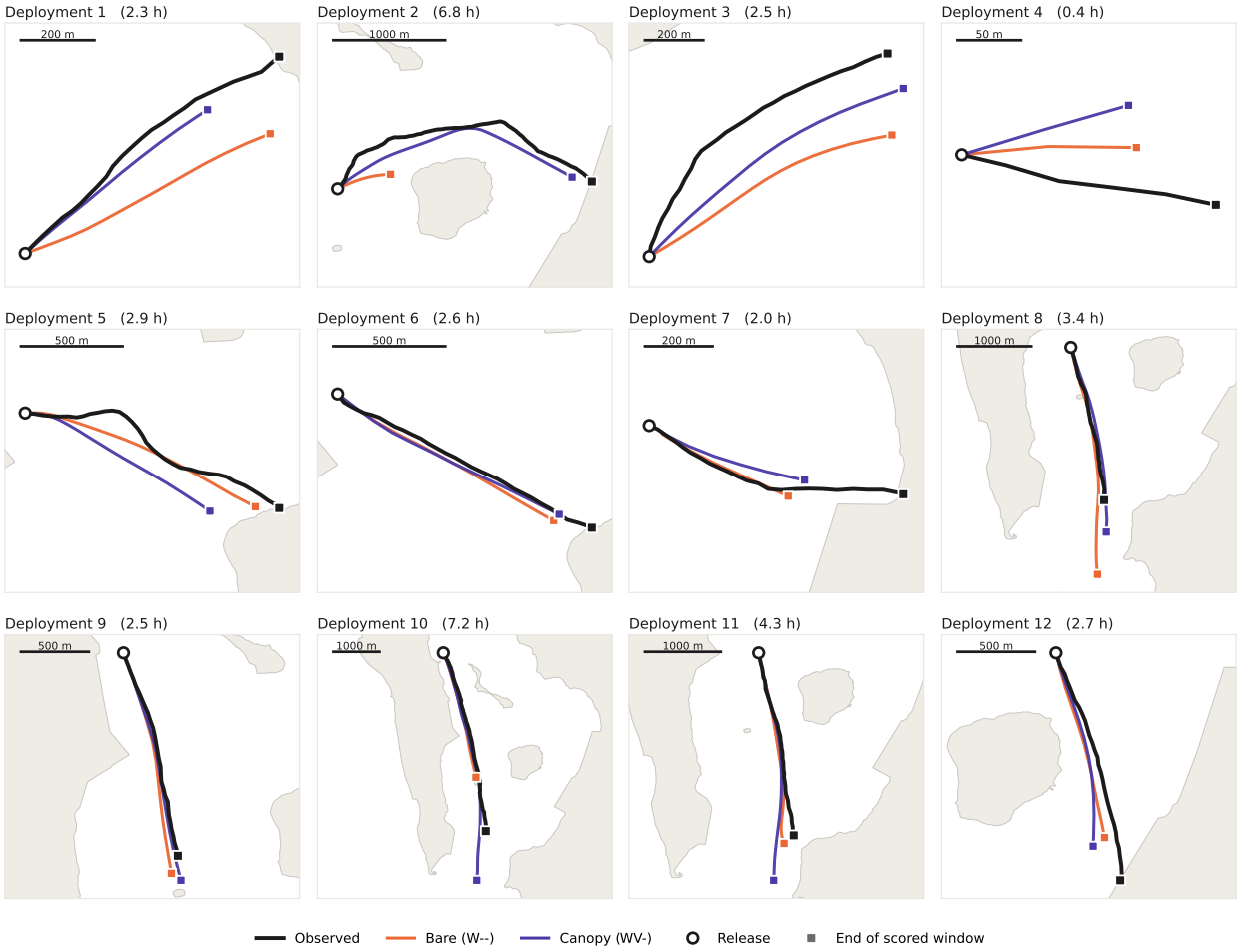


Figure 5: Observed and simulated drifter trajectories, all twelve deployments, for the wave-coupled fixed-bed configurations that differ only in the canopy. One drifter per panel, the median-skill drifter of its deployment under the vegetated configuration. Circles mark the release and squares the end of the scored window. Each track is clipped to the interval its drifter was in the water. Panel extents differ, so each carries its own scale bar.

4.3. Process attribution

On the fixed bed the design is a two-factor factorial in wave coupling and canopy, with two single-factor contrasts per factor. Figure 6 gives them, with the contrasts that involve the mobile bed beside them, with bootstrap confidence intervals over 4000 paired resamples.

Canopy. The canopy raises skill by 0.120 (95% CI +0.061 to +0.180) without waves and by 0.114 (CI +0.056 to +0.170) with them, with 25 and 27 of the 35 drifters improving. Both intervals are clear of zero, and the two contrasts agree to within 0.006, so the canopy effect does not depend on the wave treatment.

Wave coupling. Wave coupling reduces skill by 0.022 (CI −0.043 to −0.003) without a canopy and by 0.028 (CI −0.040 to −0.017) with one, with 21 and 25 of the 35 drifters worse. The effect is a fifth of that of the canopy, and the interaction term of the two factors is −0.006.

The canopy contrast holds deployment by deployment (Figure 6b), with the same sign under both wave treatments in ten of the twelve deployments. Deployment 4 is retained in all of the above and is the one deployment that argues against the canopy, its bare members scoring 0.68 and 0.69 against 0.48 and 0.50 for its vegetated ones. It is also a sampling outlier, 0.4 hours in the water over 162 m of observed path against next-lowest values of 1.9 hours and 723 m. Excluding it widens the two canopy contrasts to +0.148 and +0.144 and leaves the wave contrasts at −0.025 and −0.029.

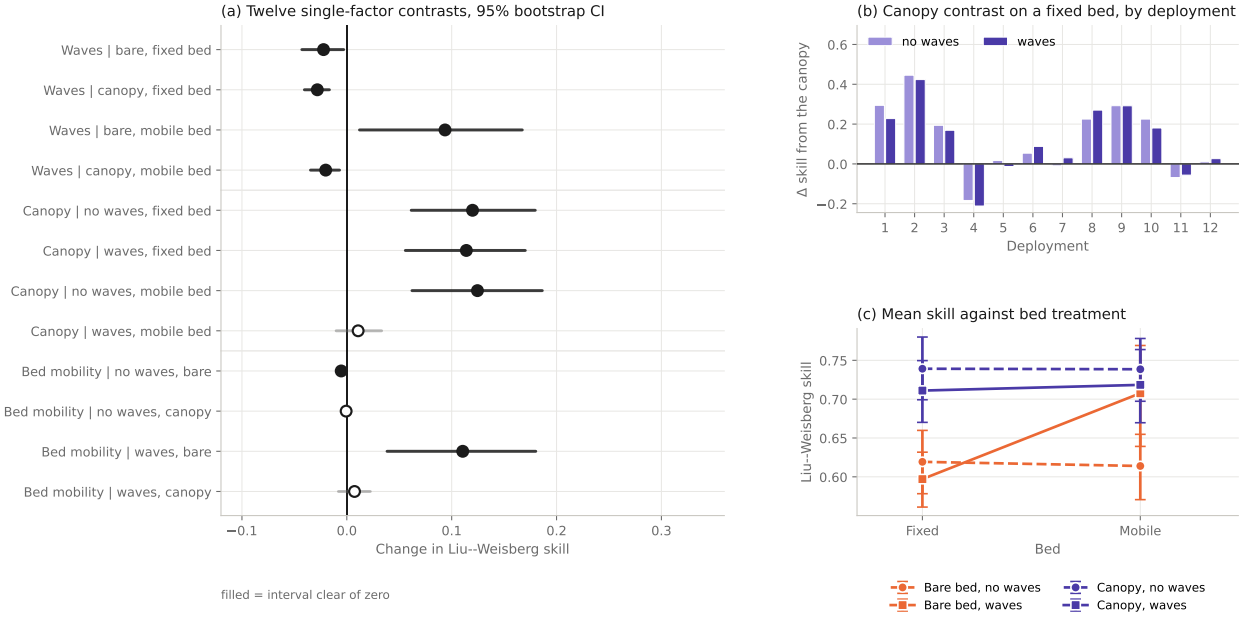


Figure 6: Process attribution. (a) The twelve single-factor contrasts of the three-factor design with 95% bootstrap intervals over 4000 paired resamples, filled markers denoting intervals clear of zero; the contrasts involving the mobile bed belong to the sensitivity arm. (b) The canopy contrast on the fixed bed, deployment by deployment, without and with wave coupling. (c) Mean skill against bed treatment, one line per combination of canopy and wave treatment, with 95% intervals on each mean.

4.4. Three-dimensional structure

Over the wet cells of the lagoon interior, with the inlets excluded, mean current speed increases monotonically from the bed to the surface in every member (Figure 7), and the canopy sets the shape of that profile. On the fixed bed without a canopy, bed-layer speed is

0.032 to 0.035 m s^{-1} and surface-layer speed 0.065 to 0.069 m s^{-1}, for surface-to-bed ratios of 1.88 and 2.14. With a canopy, bed-layer speed falls to 0.0093 to 0.0095 m s^{-1} while surface speed falls much less, to 0.043 to 0.047 m s^{-1}, and the ratio rises to 4.50 and 5.05.

Wave coupling moves the ratio by far less, adding 0.26 in the bare arm and 0.55 in the vegetated one, against a canopy contribution of +2.62 without waves and +2.91 with them.

Bed shear stress separates the two treatments by sign rather than by magnitude. In the one-day probes used to set the canopy parameters, started from the uniform column, it fell over the meadow from 0.037 to 0.0026 Pa when the canopy was introduced, a factor of 14, while a probe representing the same meadow as an equivalent bed roughness instead of as a canopy raised it to 0.049 Pa over the same cells and times.

Lagoon-mean surface speed does not order the members the way transport does, since the member reproducing the observed drifter behaviour best has the lowest lagoon-mean surface speed of the fixed-bed members at 0.043 m s^{-1}, against 0.069 for a member whose particles travel 17% too short a distance.

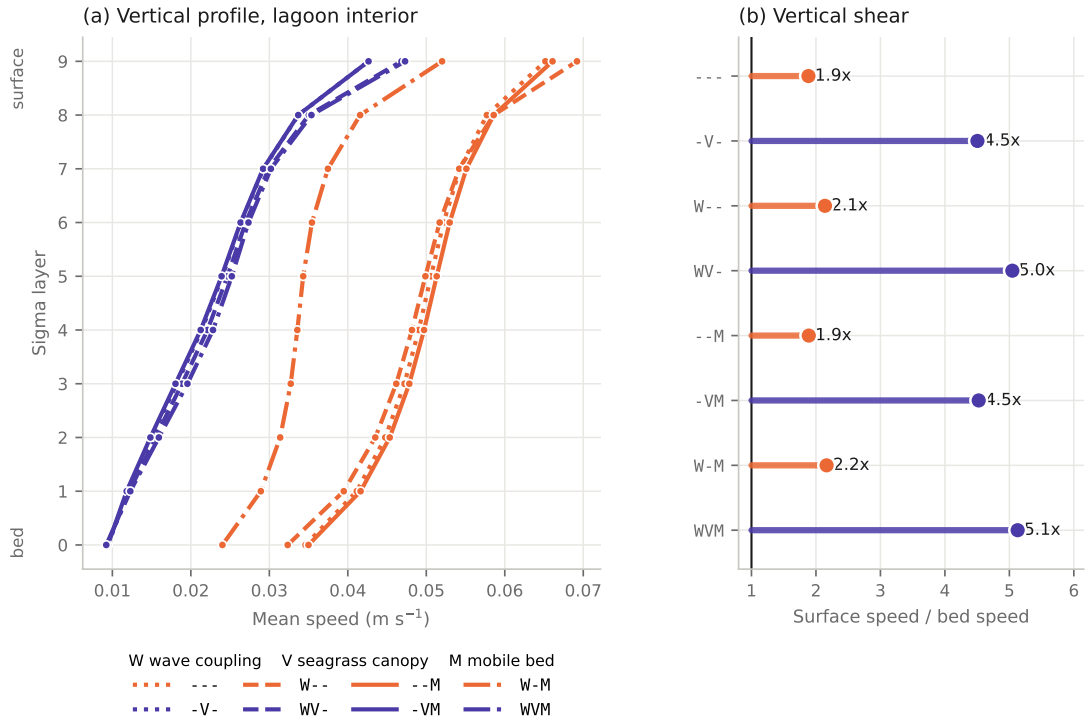


Figure 7: Vertical structure in the lagoon interior over the drifter window. (a) Mean speed by sigma layer, colour giving the canopy treatment and line style the wave and bed treatments. (b) The ratio of surface to bed speed.

4.5. Response of the seagrass classes

Cells were grouped by the class assigned to them in the seagrass map, taking the dominant class at the nearest classified point to each face centroid. Comparing classes within one configuration does not isolate the canopy, since the classes occupy different parts of the basin at different depths, and with no canopy applied at all the bare fixed-bed configurations already have their *Posidonia* cells slower at the bed than their sand cells, in a ratio of 0.91 and 0.92. The canopy effect is therefore read between configurations within a class, or as a change in that within-configuration ratio.

Under either reading the effect is large and correctly ordered, since on the fixed bed introducing the canopy reduces mean bed speed by 78% over *Posidonia* and 77% over *Cymodocea*, against 59% over sand and 57% over rock (Figure 8b). The *Posidonia* to sand bed-speed ratio falls from 0.91 and 0.92 without a canopy to 0.49 with one, under both wave treatments. The two unvegetated classes respond as well, because slowing the flow through the meadow changes the circulation of a basin whose interior is largely meadow, but they respond by a quarter less.

The canopy also removes most of the dependence on wave coupling, since within a class mean bed speed differs between the two bare fixed-bed configurations by 0.002 to 0.010 $\mathrm{m\,s^{-1}}$ and between the two vegetated ones by 0.0002 to 0.0011 $\mathrm{m\,s^{-1}}$. Once the canopy is present the near-bed flow is nearly the same whatever the wave treatment, which is the near-bed counterpart of the agreement in transport skill reported in Section 4.2.

The wave field does not follow the flow in this respect, mean orbital velocity at the bed changing by -0.4% over *Posidonia*, -0.9% over *Cymodocea*, -0.8% over sand and -0.2% over rock when the canopy is introduced on the fixed bed (Figure 8c), which is neither large nor ordered by canopy density. The reason is structural rather than numerical. In this coupling the flow model passes a bed roughness field to the wave model, and the canopy is represented in the momentum equation of the flow rather than in that roughness field, so it never reaches the wave computation. The seagrass in these simulations attenuates the current and not the waves, and Section 5.4 returns to what that leaves unresolved.

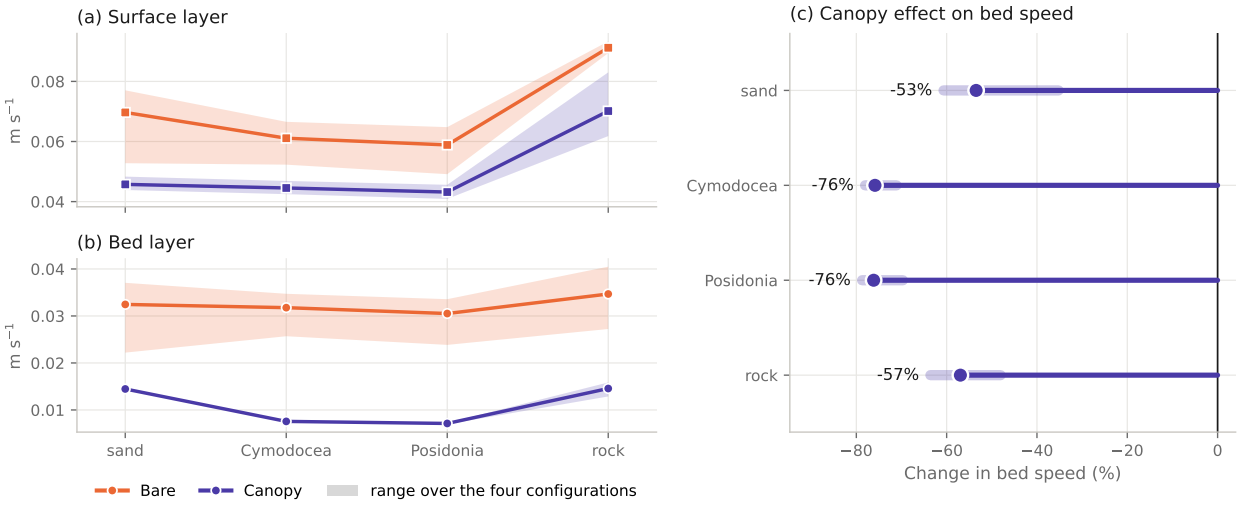


Figure 8: Flow response by seagrass class in the lagoon interior. (a) Mean speed in the surface layer and (b) in the bed layer, the line giving the mean over the four configurations of each arm, the mobile-bed ones included, and the band their range. (c) Change in mean bed speed on introducing the canopy, the marker giving the mean over the four cells of the design and the band their range.

4.6. Sensitivity to bed mobility, initial condition and discretisation

With wave coupling the mobile bed leaves any plausible state of the basin. Averaged over the lagoon from 7 to 10 July, suspended sediment reaches 9.3 $\mathrm{g\,L^{-1}}$ in the bare wave-coupled member and 9.4 $\mathrm{g\,L^{-1}}$ in its vegetated counterpart, against 0.1 and 0.007 $\mathrm{g\,L^{-1}}$ in the two uncoupled mobile-bed members, and the bed of the lagoon loses 4.0 and 3.1 million $\mathrm{m^3}$ over the nine days. By 10 July the bare wave-coupled member has lowered the lagoon bed by 12 cm on average and changed it by more than 5 cm over 74% of its area, whereas without waves the change exceeds 5 cm over 0.3% of the area. The canopy does not shield the bed from this erosion, because it does not reach the wave model.

The same bare member nevertheless scores 0.708 (Table 3), close to the vegetated members, so that in the cell combining waves and a mobile bed the canopy contrast falls to +0.011 (CI −0.010 to +0.033) and the wave and bed-mobility contrasts of the bare arm reach +0.094 and +0.111 (Figure 6a). The two segments restarted on 7 July locate the source of that skill. With the bed updated the segment scores 0.710 and with the bed held at its 7 July state 0.707, a difference of -0.003 ± 0.004, while the held bed scores 0.110 ± 0.039 above the fixed-bed member `W--`, with 29 of 35 drifters better. The gain therefore comes from the bathymetry reshaped over the first six days and not from sediment transport in the drifter

window. At AltaVilaEst the member keeps an anomaly RMSE of 0.040 m and a correlation of 0.86, against 0.024 m and 0.94 for its vegetated counterpart.

Started from a vertically uniform column instead of the stratified state, the bare members score 0.080 to 0.145 lower and the vegetated members 0.005 to 0.018 lower, so that the canopy contrasts on the fixed bed become +0.195 and +0.188. The uniform start raises the modelled mean water level at the gauges by 2 to 10 cm and, on the fixed bed, changes anomaly RMSE by at most 4 mm and correlation by at most 0.03, and the canopy signature at AltaVilaEst is the same under both starts.

On the uncoupled fixed-bed configuration from the uniform start, Liu–Weisberg skill over the 35 drifters varied within 0.553 ± 0.004 across the three quadtree meshes without a monotonic trend, since the paired differences between consecutive levels, $+0.005$ and -0.007 with a standard error of 0.002, change sign, and the present mesh carrying the same bed scored 0.555, inside that band. Water level at the three gauges changed by less than 1 mm in bias and by at most 1.5 mm in anomaly RMSE between 60 and 15 m. From the stratified start, skill was 0.619 at both 10 and 20 layers, a paired difference of $+0.0003 \pm 0.0036$, and water level at the gauges differed by at most 2.4 mm in bias and 1.6 mm in anomaly RMSE. Limiting the time step also by the vertical fluxes changed skill at 10 layers by $+0.0007 \pm 0.0007$ and water level by less than 0.3 mm. Neither the horizontal nor the vertical resolution of the present configuration therefore limits the quantities the study reports.

Under the Courant limit of 0.7 used by the ensemble, the uncoupled member started from the uniform column scored 0.034 ± 0.004 lower on four partitions on the second computer than on eight on the server, with all 35 drifters worse. Under a limit of 0.5 the same comparison gave $+0.002 \pm 0.001$ on the present mesh and no difference on the 60 m quadtree mesh.

5. Discussion

5.1. Vertical placement of the canopy drag

This study asked which modelled processes control transport in a shallow vegetated lagoon, and on the fixed bed the answer the ensemble gives is the seagrass canopy, with an effect five times that of wave coupling and the same under both wave treatments. The mechanism the model offers for that is vertical rather than frictional, since introducing the canopy removes

momentum from the lower part of the water column without removing a proportionate amount from the upper part, more than doubling the ratio of surface to bed speed. Such a profile is the signature of a canopy shear layer rather than of a rougher bed, which acts through a logarithmic profile whose surface-to-bed ratio is bounded well below the value obtained here (Nepf, 2012), and the bed shear stress confirms the distinction by its sign, falling under the canopy where an equivalent bed roughness raises it over the same cells. Since a surface drifter samples the top of that profile, a treatment that reshapes the profile relocates the drifter even where it changes the depth-averaged transport little.

That places the present result in a particular relation to the two earlier models of this basin rather than in opposition to them. Ingrassia et al. (2024) obtained a best-fit uniform Gauckler–Strickler coefficient for the vegetated lagoon within a depth-averaged framework, where an equivalent bed roughness is the appropriate representation of a canopy because there is no vertical profile for the canopy to shape. De Marchis et al. (2012) worked in three dimensions and assigned Nikuradse roughness heights by seagrass class, establishing the vertical structure of the wind- and tide-driven circulation that motivates using a three-dimensional solver here at all. Both answered the questions they set, and a roughness representation reproduces the drag the meadow exerts. What has changed since is the availability of a canopy formulation that resolves that drag over the vertical extent the plants occupy, and the contribution of the present work is to quantify what that capability buys in a basin of this depth. The canopy parameters used here were taken from published measurement of the two species and were not adjusted against any observation, so the improvement follows from where the drag is applied rather than from how it was tuned.

5.2. Information carried by each gauge position

Water level separated the configurations by the canopy at AltaVilaEst, by wave coupling at BocaNord and by neither at BocaSud, and the three positions constrain different parts of the problem. Agreement at the inlet gauges establishes that the offshore forcing, the datum and the exchange through the two mouths are right, and in a micro-tidal basin with two open connections those set most of the level there, the wave contribution to the northern inlet being the one dynamic effect that reaches it. Agreement in the interior tests something the outer pair cannot, because the level there is set by what the water does on its way in, which

is where the canopy acts. The roles are complementary, and a match in the interior would be difficult to interpret without the exchange having been established first.

Together with the drifter result, the gauge result refines the familiar argument that water level constrains transport weakly and that Lagrangian observations form a distinct constraint rather than a confirmation (Liu and Weisberg, 2011; Révelard et al., 2021). Water level is not inherently insensitive to the canopy, since the one gauge sited inside the vegetated basin identified the same factor as the drifters and ranked the fixed-bed configurations the same way. For a basin of this type the useful question is therefore less whether to validate on water level than where to place the instrument intended to discriminate, and the interior of the vegetated compartment is a considerably more informative position for that purpose than the inlet. Section 5.3 shows that the interior gauge also discriminates where the drifters do not.

5.3. Bare-bed skill through a reshaped bathymetry

The mobile-bed arm contains one configuration that, taken on its own, supports a strong statement about process coupling. In the bare arm, wave coupling and bed mobility each leave Lagrangian skill nearly unchanged or lower it when applied alone and raise it when applied together, with an interaction term of +0.116, tight bootstrap intervals and a mechanism available in the literature, since wave orbital motion sets the bed stress that mobilises sediment and wave–tide coupling is known to govern the sediment dynamics of shallow micro-tidal basins (Carniello et al., 2011). The combination brings the bare member to 0.708, within 0.03 of the vegetated members, and removes the canopy contrast in that cell.

The segments restarted on 7 July show that the statement has no physical basis in this model. The skill is carried by the bathymetry that six days of wave-driven erosion reshaped, through a lagoon holding near 9 $\mathrm{g\,L^{-1}}$ of suspended sediment and losing 4 million $\mathrm{m^3}$ of bed in nine days, a state the basin does not reach in summer. The reshaped bed does not reproduce the vertical structure of the canopy either, the surface-to-bed speed ratio of that member remaining at 2.17 against 5.13 with the canopy, so the agreement in skill is reached through a different flow. A model without a canopy can therefore reach the drifter skill of a model with one through a mechanism that is wrong, and the drifters alone do not separate the two. The interior gauge does, since at AltaVilaEst the bare member keeps an anomaly RMSE of 0.040 m and a correlation of 0.86 against 0.024 m and 0.94 for its vegetated counterpart.

The initial condition shows the same asymmetry by another route. A start from a uniform offshore column lowers the skill of the bare members by 0.08 to 0.15 and that of the vegetated members by at most 0.02, so the canopy contrast grows from about 0.12 to about 0.19 when the start is wrong. In both cases an error elsewhere in the model degrades the bare configurations and leaves the vegetated ones nearly unaffected, and the magnitude of a canopy contrast therefore depends on what else the model gets wrong, even where its sign does not.

What follows from this bears on how such an ensemble is read rather than on this basin alone, since canopy drag is commonly left out of lagoon and estuary models or folded into a bed roughness coefficient, and an ensemble built that way can still return a well-defined and statistically clean interaction between the processes it does carry. Here that interaction is produced by an implausible morphodynamic state, and nothing internal to the bare half of the design distinguishes it from a physical result. What distinguished them was varying the canopy as a factor in its own right, checking the morphodynamic state against a plausible sediment budget, and scoring the result against two observation types that no model decision had used.

5.4. Limitations

Wave dissipation by the canopy. The canopy does not reach the wave model, because the coupling passes a bed roughness field to the spectral model while the canopy enters the momentum equation of the flow, so mean orbital velocity at the bed changes by under 1% when the canopy is introduced and not in an order that follows canopy density. Seagrass meadows do attenuate waves, in this species and at this scale (Infantes et al., 2012; Prinos et al., 2010), and shallow restricted Mediterranean lagoons carry wave fields that respond to bed conditions (Morote-Sánchez et al., 2025). The wave field in these simulations is therefore that of a bare bed, the wave contrasts reported here are conditional on that, and so is the wave-driven erosion that the canopy fails to shield in the mobile-bed arm. Representing canopy dissipation inside the spectral model is the corresponding extension. It would be expected to change the wave contrasts rather than the canopy ones, which are measured on the flow.

Canopy reconfiguration. The canopy bends under load rather than holding a fixed height, and the response is dynamic, the velocity ratio between two probes differing only in

leaf stiffness moving from 1.005 in the slowest quintile of the record to 0.807 in the fastest 5%, where a fixed shortening would give a constant ratio. A 13% change in leaf density moves the velocity field by 20% at the upper end of the range, which is a strong sensitivity to a value taken from a published table rather than measured in this meadow. The bending is moreover mechanical, whereas the flume measurements of Ciraolo et al. (2006b) describe reconfiguration empirically through a friction factor that varies by a factor of twelve over the velocity range they tested. That empirical formulation is available in the model only in its depth-averaged solver, so the two could not be compared directly and the agreement between them was not tested.

Bed mobility. The morphodynamic arm is not calibrated, the critical shear stress for erosion having been constrained by neither a repeated bathymetric survey nor a turbidity record, and with waves it leaves any plausible state of the basin, with suspended sediment near 9 $\mathrm{g\,L^{-1}}$ and the lagoon bed lowered by 12 cm on average in nine days (Section 4.6). It is therefore a sensitivity test and is kept out of the attribution. Before it could enter it, the erosion parameters would need to bring suspended sediment in the lagoon to the order of tens of milligrams per litre.

Window and restart segments. The record covers two days of water level and thirty-four hours of drifter observations within one summer season, and the vegetated configurations are three-day restart segments. A segment run without any other change reproduces its continuous member in water level and drifter skill (Section 4.1), but seasonal behaviour, storm conditions and the interaction of a canopy with a mobile bed over longer periods are outside what this design addresses.

Initial condition. The ensemble starts from the depth profiles of the CMEMS field at the start time, which a start from a uniform column does not approach within the nine days, and from the uniform start the canopy contrasts on the fixed bed are +0.195 and +0.188 rather than +0.120 and +0.114 (Section 4.6). The stratified start is itself a single profile applied over the domain and leaves out the horizontal structure of the CMEMS field, so the canopy contrasts reported here may still carry part of a start-up adjustment. The solution also depends on the host and the decomposition at the Courant limit of the ensemble, by 0.034 in skill from the uniform start, which all members share, while the horizontal and vertical

resolution do not limit the reported quantities.

***Cymodocea nodosa* parameters.** The *C. nodosa* parameters are loosely constrained, coming from a flume study of plants collected elsewhere, since no measurement of that species in this lagoon was available. The species carries about 1.5% of the canopy frontal area in the model, which bounds the consequence, and the drag is carried almost entirely by the *P. oceanica* column.

6. Conclusions

In a shallow Mediterranean lagoon where the seagrass canopy occupies most of the basin and the mean depth is below one metre, the canopy emerges from this ensemble as the first-order control on Lagrangian transport. On the fixed bed, representing it raises Liu–Weisberg skill by 0.120 without wave coupling and by 0.114 with it, both bootstrap intervals clear of zero, against -0.022 and -0.028 for wave coupling, and the two factors do not interact.

The results indicate that where the drag is applied matters more than how it is calibrated. The canopy here is a three-dimensional drag field parameterised entirely from published measurement of the two species, with nothing fitted, and it more than doubles the ratio of surface to bed speed while lowering bed shear stress by a factor of 14, where representing the same meadow as an equivalent bed roughness raises that stress. A treatment that changes the shape of the velocity profile relocates a surface drifter, where a treatment changing only its magnitude does not.

It follows that process attribution from a factorial is conditional on the factors the factorial contains and on the state the model reaches. With waves, the uncalibrated mobile bed brings a configuration without a canopy to 0.708, within 0.03 of the vegetated ones, and produces an interaction of $+0.116$ between waves and bed mobility. The skill is carried by a bathymetry that six days of erosion reshaped while the lagoon held near 9 $\mathrm{g\,L^{-1}}$ of suspended sediment, and holding the bed fixed over the drifter window leaves it unchanged. A start from a uniform offshore column likewise degrades the bare configurations and leaves the vegetated ones nearly unaffected, raising the canopy contrasts to 0.19. No statistic internal to the bare configurations reveals either compensation.

The same design shows that where a water level gauge sits decides what it can constrain.

Anomaly RMSE differs across the four fixed-bed configurations by 1.6 mm at the southern inlet, by 12 mm at the northern inlet, where it follows wave coupling, and by 32 mm at the gauge inside the vegetated basin, where it follows the canopy and correlation moves from 0.79–0.81 to 0.94–0.95. The interior gauge also separates the canopy treatments in the configuration where the drifters do not, the bare wave-coupled mobile-bed member keeping an anomaly RMSE of 0.040 m against 0.024 m with the canopy.

The canopy is not a fixed property of this basin, its extent having fallen substantially over the past two decades, and the two arms of the design bracket what a loss of that kind does to the flow, one carrying the meadow as mapped and the other none of it. Between those two states mean bed speed over the meadow changes by 78%, the surface layer moves from about twice the speed of the bed to about five times, and Lagrangian skill separates without overlap. The transport regime of a basin of this type is therefore a property of its meadow as much as of its geometry and its forcing.

CRediT authorship contribution statement

Martins Jr, C.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **De Marchis, M.:** Conceptualization, Supervision, Methodology, Writing – review & editing. **Ciraolo, G.:** Conceptualization, Supervision, Methodology, Writing – review & editing. **Nasello, C.:** Conceptualization, Supervision, Methodology, Writing – review & editing. **Maltese, A.:** Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Software and data availability

- **Model:** Delft3D Flexible Mesh Suite 2026.01 HMWQ (D-Flow FM kernel 1.2.184) coupled online with SWAN through DIMR.
- **Source code repository:** `https://github.com/cicero-martins/StagnoneDT`
- **Data availability:** ERA5 and CMEMS products are openly available from the Copernicus programme. PlanetScope imagery is under commercial licence, and scene identifiers are listed in the repository for traceability. In-situ water level, wind, and drifter records are archived in the repository.

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